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1. CM5 RP1 Secondary SDIO (<=12.5 MHz Default-Speed Mode, Type-B 1:1 FFC & Dual-OR Return Interleaving): Bottom-mounted J_SUB_FIO (B.Cu mirrored pad X-coordinates: Pin 1 at +X=80.25mm, Pin 22 at -X=69.75mm) mates via a Type-B (opposite-side contact, 1:1 pinout) 22-pin 0.5mm FFC cable to top-mounted carrier J_FIO (F.Cu), with J_SUB_FIO.8 and J_SUB_FIO.13 tied to GND via dedicated parallel 0R resistors R28 and R36. Constrained to <=12.5 MHz (80 ns period via CM5 RP1 mmc DT overlay max-frequency = <12500000>) with 33R host-side series resistors at U_CM5, 22R card-side series damping/ESD-isolation resistors R40, R12, R14, R15, R37, R38 (0402WGF220JTCE, C25092) on CLK, CMD, and DAT0..3 between U7.B1..B6 and U2/U3/J_SD, and 47k card-side pull-ups R1-R5 (47k || 100k = 32.0k).
2. Constant-Current Auto-Retry High-Side SD Power Switch (U1 AP22653W6-7, C2158037) & Self-Held Active /SD_VDD Discharge (Q1, Q4, R39, D2, D3, C18): Controlled by SD_PWR_ON (R8=10k + C15=100pF C1546 HF filter keeping GPIO5 capacitive load <=135pF. R22=56k 1% C25796 setting ILIM_min=365.5mA, ILIM_typ=420.6mA, ILIM_max=483.5mA). Dedicated N-MOSFET inverter Q4 (BSS138 C78284, G=SD_PWR_ON, S=GND, D=SD_PWR_OFF) with R39=10k pull-up to /SD_DIS_VCC (OR-ed from /3V3_CM5 via D2 RB751V40T1G and /SD_VDD via D3 RB751V40T1G onto C18=100nF hold-up cap) keeps Q1 strictly OFF when SD_PWR_ON=HIGH and holds Q1 gate at ~3.0V when SD_PWR_ON=LOW even if /3V3_CM5 collapses to 0V, discharging 20.1uF below 0.5V in 5.96ms.
3. Direct /SD_VDD-Sensed (<150ns Brownout/Short & <20ns Disable/Fault Turn-Off) + Turn-On Delayed (>=3.05ms min / 4.72ms typ) 6-Channel SDIO Bus Isolation (U7 SN74CB3Q3245PWR + U8_3 + U9 + Q3/Q2 + D4/D5/D6): RC delay network R35=56k (from /SD_VDD) + C14=100nF (with fast /SD_VDD brownout reset diode D4 RB751V40T1G and FAULT reset diode D5 RB751V40T1G upstream of U8.6 3A -> 3Y SD_PWR_DLY_BUF) feeds Schmitt-trigger 2-input AND gate U9 (SN74LVC1G97DBVR C128411, IN0=SD_PWR_ON, IN2=SD_PWR_DLY_BUF -> Y=SD_PWR_EN_AND) gating Q3 in series with Q2 (gated directly from /SD_VDD via R31=1k + C16=100pF C1546 tau=127ns and pulled low on FAULT via D6 RB751V40T1G, with R33=1k to /SD_VDD) so U7 opens in <150ns on /SD_VDD short/brownout without waiting for U1 t_Blank and always waits the full R35/C14 soft-start delay after brownout or fault recovery.
4. Non-Snapback 3.3V TVS on /SD_VDD (D1 PTVSLC3D3V3B, C246774), Inline Flow-Through 3.3V Snapback Signal ESD (U2, U3, U4, U6 RCLAMP3324P.TNT with opposite flow-through pin pairs 1-10, 2-9, 4-7, 5-6 connected inline and /SD_CMD assigned to U3) & Schmitt-Trigger Switch Conditioning (U5, U8).

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Title: Compact all-in-one CM5 desktop computer - Front I/O Sub-board

Size: A2	Date:	Rev:
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